Regularized Logistic Regression

Just like linear regression, logistic regression can be regularized to prevent overfitting, especially when using a large number of features or high-order polynomial terms.

The Problem: Overfitting in Logistic Regression

When logistic regression is trained with many features, it can create an overly complex decision boundary that fits the training data perfectly but fails to generalize to new data.

The Regularized Cost Function

To address overfitting, we add a regularization term to the logistic regression cost function. This penalizes large values for the weight parameters (w_j).

- **The final cost function for regularized logistic regression is:**
  $J(\vec{w}, b) = \left[-\frac{1}{m} \sum_{i=1}^{m} y^{(i)} \log\left(f_{\vec{w},b}(\vec{x}^{(i)})\right) + (1 - y^{(i)}) \log\left(1 - f_{\vec{w},b}(\vec{x}^{(i)})\right)\right] + \frac{\lambda}{2m} \sum_{j=1}^{n} w_j^2$
- Adding this term encourages the algorithm to find smaller parameter values, resulting in a simpler, smoother decision boundary that is less likely to overfit.

Gradient Descent Implementation 👨‍💻

The gradient descent algorithm is updated to account for the new regularization term in the cost function.

- **Partial Derivatives:**
  - The derivative with respect to w_j gets an additional term:
    $\frac{\partial J}{\partial w_j} = \left(\frac{1}{m} \sum_{i=1}^{m} (f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}) x_j^{(i)}\right) + \frac{\lambda}{m} w_j$
  - The derivative with respect to b remains **unchanged**, as the bias term is not regularized.
- **Simultaneous Update Rules:**
  - **Update for w_j:**
    $w_j := w_j - \alpha \left[\left(\frac{1}{m} \sum_{i=1}^{m} (f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}) x_j^{(i)}\right) + \frac{\lambda}{m} w_j\right]$
  - **Update for b:**
    $b := b - \alpha \left[\frac{1}{m} \sum_{i=1}^{m} (f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)})\right]$
- **Key Similarity:** The update rule equations are **identical** to those for regularized linear regression. The only difference is the definition of $f_{\vec{w},b}(\vec{x})$, which is the sigmoid function for logistic regression.